TRACE FORCE MRV

Cement Embedded Emissions — Internal QA/QC Working Report

Reporting & Operational Parameters

Installation / Site:	Demo Cement Plant - Egypt
Country / Jurisdiction:	Egypt
Primary Product:	CEM II/A-L 42.5N
Reporting Quarter:	Q1 2025
Accounting Period:	2025-01-01 to 2025-03-31
Data Origin:	workbook
QA/QC Overall Status:	INCOMPLETE
Required Data Completeness:	100%
Internal Review Readiness:	Not ready
Evidence Mapping Status:	Mapped (not externally verified)
External Verification:	None recorded — not externally verified
Report Generation Date:	2026-10-19 10:31:51 UTC

Scope & Limitations Statement

Internal QA/QC working report. Figures are computed by the platform from the supplied inputs. This report has NOT been externally verified, assured or approved, and makes no claim of regulatory compliance. "Mapped" evidence means only that a document reference is linked in the evidence register.

Emissions & Production Summary

Computed by the platform calculation engine from the supplied inputs. Missing inputs show as 'not calculated'.

METRIC / INDICATOR	COMPUTED VALUE	UNIT
Clinker Production	**520,000**	t
Cement Production	**680,000**	t
Clinker Factor	**0.765**	ratio
Process CO2 (Selected)	**276,900**	tCO2
Fuel Combustion CO2	**187,378.0559**	tCO2
Biogenic CO2 Memo	**8,278.3207**	tCO2
Direct Embedded CO2	**464,278.0559**	tCO2
Grid Electricity Consumed	**95,000**	MWh
Indirect Grid CO2	**46,341**	tCO2
Total Embedded CO2	**510,619.0559**	tCO2
SEE Clinker	**0.8928**	tCO2/t clinker
SEE Cement	**0.7509**	tCO2/t cement
Specific Heat Consumption	**3.9892**	GJ/t clinker
Thermal Substitution Rate	**11.8098**	%

Automated QA/QC Check Register

17 of 18 checks PASS · 1 not evaluated (missing inputs) · QA pass rate 94% · completeness 100%

ID	CATEGORY	CHECK NAME	RESULT	STATUS
MB-01	Mass Balance	Clinker factor range	0.765	PASS
MB-02	Mass Balance	Cement production >= clinker used	OK	PASS
MB-03	Mass Balance	Cement mass balance tolerance	0.00%	PASS
EI-01	Emission Intensi	SEE clinker expected range	0.8928	PASS
EI-02	Emission Intensi	SEE cement not extreme	0.7509	PASS
FC-01	Fuel	All fuel NCV populated	17	PASS
FC-02	Fuel	All fuel EF populated	85	PASS
FC-03	Fuel	Biomass fraction valid	0 to 0.5	PASS
FC-04	Fuel	Oxidation factor reasonable	0.98 to 0.99	PASS
EL-01	Electricity	Grid EF populated	0.4878	PASS
EL-02	Electricity	Electricity present	95000	PASS
CP-01	Completeness	Process CO2 calculated	276900	PASS
EV-01	Evidence	Evidence register linkage	0	PASS
PR-01	Period Lock	Input period aligned	0 outside, 0 undat	PASS
MB-04	Mass Balance	Clinker stockpile reconciliation	no inventory	NOT_EVALUATED
EI-03	Emission Intensi	SHC expected range	3.9892	PASS
FC-05	Fuel	TSR in expected range	11.8098	PASS
CP-02	Completeness	Method selection valid	Method B - clinker	PASS

Evidence Register & Regulatory References

15 of 15 evidence references mapped in the register (mapping only — not externally verified).

EV-PROD-JAN	January clinker/cement production totals	Mapped
EV-PROD-FEB	February clinker/cement production totals	Mapped
EV-PROD-MAR	March clinker/cement production totals	Mapped
EV-RAW-JAN	January carbonate chemistry / raw meal lab	Mapped
EV-RAW-FEB	February carbonate chemistry / raw meal lab	Mapped
EV-RAW-MAR	March carbonate chemistry / raw meal lab	Mapped
EV-FUEL-PETCOKE-Q1	Petcoke invoices, weigh feeder, lab NCV	Mapped
EV-FUEL-COAL-Q1	Coal invoices, weigh feeder, lab NCV	Mapped
EV-FUEL-RDF-Q1	RDF weigh feeder and biomass fraction evidence	Mapped
EV-FUEL-TYRE-Q1	Tyre chips quantity and biomass fraction evidence	Mapped
EV-ELEC-JAN	January purchased grid electricity	Mapped
EV-ELEC-FEB	February purchased grid electricity	Mapped
EV-ELEC-MAR	March purchased grid electricity	Mapped
EV-FACTOR-LIB	NCV, EF, oxidation factor and grid EF governance	Mapped
EV-QAQC-Q1	Quarter-level QA/QC review pack	Mapped

Reference Standards (for context only)

CBAM Regulation (EU) 2023/956
Legal basis for CBAM and covered goods / embedded emissions reporting.

Commission Implementing Regulation (EU) 2023/1773
Transitional phase reporting rules and annexes.

CBAM guidance for installation operators outside the EU
Operator-facing guidance for installation-level data and communication with importers.

CBAM Communication Template for installations
Template designed to help operators transmit data correctly.

CBAM Communication Template examples
Sector examples include cement, aluminium, fertilisers, hydrogen and steel.

Default values transitional period
Reference when actual data is unavailable for CBAM-covered goods.

EU ETS MRR concepts
Monitoring and reporting discipline for emissions data and factors.

ISO 14064-1:2018
GHG inventory boundaries and quantified emission categories.

ISO 14064-3:2019
Validation/verification concepts for GHG assertions.